

Worksheet 19

Understanding slip

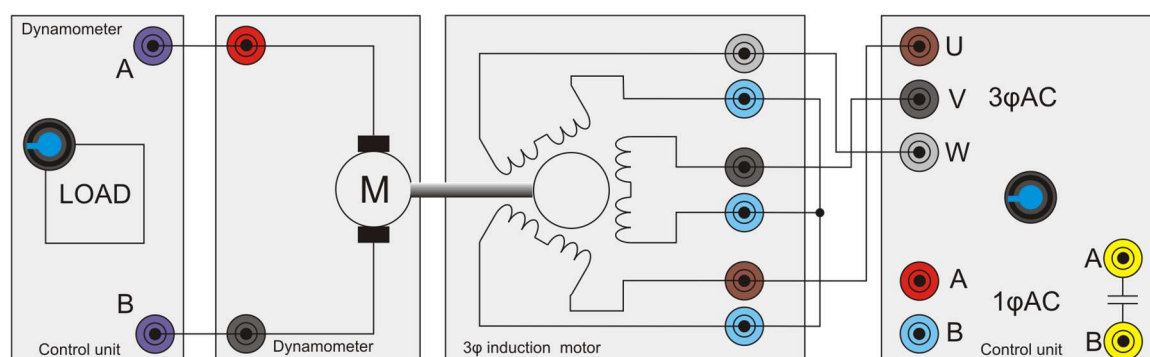
Modern electrical machines



Whilst induction motors may vary in size, their characteristics are all similar.

Larger motors can be very expensive. When they fail, they are refurbished by hand - a specialist task.

In the photograph, an engineer is re-winding an induction motor.



Over to you:

- 1) Set up the three-phase induction motor as shown in the diagram above.
- 2) On your PC run 'Open Control_3Phase.bat'.
- 3) Calculate the synchronous speed for the three-phase motor using the formula in the 'Key relationships' section.
(Refer to the cross-section drawings in the 'Reference' section to see how many poles this motor has.)
- 4) With no dynamometer load, for drive frequencies of 40Hz, 50Hz, 60Hz, 70Hz and 80Hz, calculate the slip at each frequency. Record the results in the table opposite.
- 5) Set the drive frequency to 50Hz. Increase the dynamometer load to 50%, then 75%, then 90% and finally full load. Calculate the slip at each step. Record the results in the table.

3p motor drive	Speed	Slip
40Hz		
50Hz		
60Hz		
70Hz		
80Hz		

Slip vs drive frequency - no load

Dyno load	Speed	Slip
0%		
50%		
75%		
90%		
full		

Slip vs load - drive frequency = 50%

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Over to you:

Explain these readings.

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.